McMaster University

Bachelor of Technology Program

SFWRTECH 4WP3 – Advanced Web Programming

# Project #2 - Proposal

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# Lake Ledger

## Users and Purpose:

For avid paddlers, it can become difficult to keep track of all the routes they have completed. Keeping a log of past trips can help when planning future trips or sharing experiences with others. Lake Ledger is a web application where paddlers can log their canoe trips and manage a collection of route objects using CRUD operations.

## API Routes:

**GET** /api – retrieve entire collection of canoe trips
This route will be triggered when the app is opened, refreshed, or when a "View All Trips" button is pressed. The trips will be displayed in a scrollable list within the application showing the route name, distance, and difficulty rating.

**POST** /api – creates a new canoe trip to be logged
This route will be triggered when a user fills out a form with trip details and presses an "Add Route" button. The new trip will then be inserted into the database.

**DELETE** /api – deletes the entire collection of canoe routes
This route will be triggered when the user selects a "Clear All Trips" button, removing every route currently stored in the database.

**GET** /api/id – retrieve a specific canoe route from using its ID
This route will be executed when a user selects a route from the list to view its details. The application will request the specific route using its unique ID and display its full information.

**PUT** /api/id – used to replace a specific item with a new entry
This route will be used when a user edits an existing canoe trip. The application will send updated information to the API, replacing the existing entry with the new details.

**DELETE** /api/id – delete a specific object
This route will be used when a user presses a "Delete" button next to a trip. The selected route will then be removed from the database.

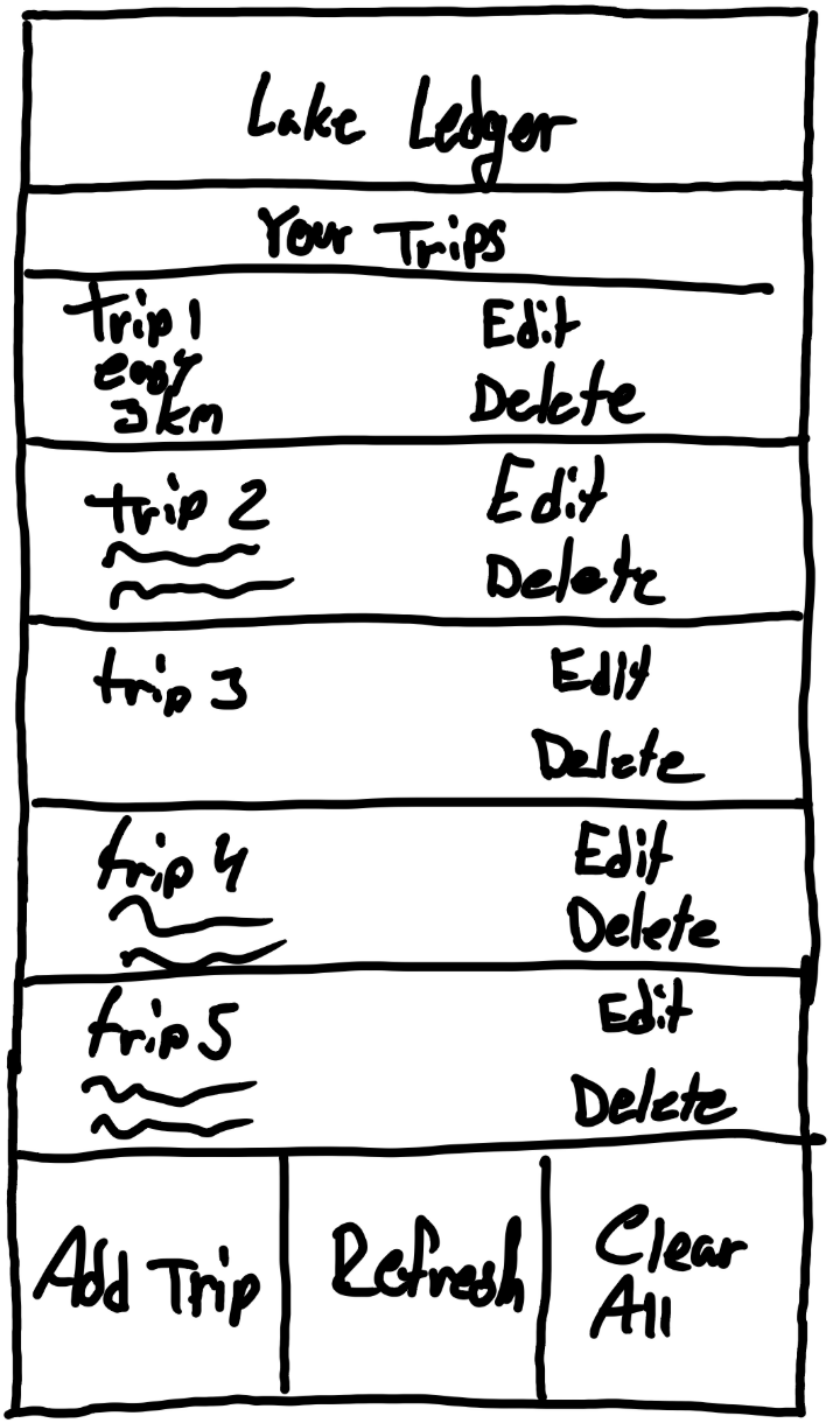
## Fields and Types:

id: INTEGER

routeName: TEXT

difficulty: INTEGER

distance: REAL (for decimals in total route KMs)

description: TEXT

UI Prototype: